ML16 Predicting a category with a decision tree

Week 6, Trees and fair comparisons | Session 16 of 24 | Machine Learning | New library: sklearn

Today's question. Does a tree that learned its questions beat my hand-written flower rule?

Objective. By the end: you can train a small decision tree on iris, read its first question, and compare its accuracy with a baseline and with the hand rule from session 4.

> **WHERE THIS IS USED**
> Doctors and farm advisers sort cases into groups by asking questions in order: is there a fever, are the leaves spotted? A decision tree learns which questions to ask. Can your tree out-guess your own rule?

Before this session: ML15 A complete regression experiment. If you cannot explain the last example, use its error card with the instructor first.

Words for today

Word	What it means	Think of it like this
classification	Predicting a category instead of a number.	Deciding whether a fruit is a mango or an orange.
decision tree	A chain of yes or no questions about the features that ends in a category.	A flow chart that sorts the flowers.
class label	The name of a category; its code number is only an identifier.	The species name on a seed packet.
accuracy	The share of predictions that are right.	The shots that go in, out of all the shots.
majority baseline	A model that always names the most common category.	Always answering the most popular option.
depth	How many questions in a row the tree may ask.	How many forks a path may take.

Block A. Train a tree and draw it

Split the 150 iris flowers, train a tree that may ask two questions in a row, and draw it.

Predict. Predict whether the tree's accuracy will be above 85% and whether the majority baseline will be above 55%. Write your guess on paper first.

Your guess, written before you run it

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.dummy import DummyClassifier
from sklearn.metrics import accuracy_score
```

```
import matplotlib.pyplot as plt
iris = load_iris()
X_train, X_test, y_train, y_test = train_test_split(iris.data, iris.target,
    test_size=0.25, random_state=7)
tree = DecisionTreeClassifier(max_depth=2, random_state=7).fit(X_train, y_train)
baseline = DummyClassifier(strategy="most_frequent").fit(X_train, y_train)
tree_acc = accuracy_score(y_test, tree.predict(X_test))
base_acc = accuracy_score(y_test, baseline.predict(X_test))
print(round(tree_acc, 3), round(base_acc, 3))
plot_tree(tree, feature_names=iris.feature_names, class_names=list(iris.target_names))
plt.show()
```

Run it, then compare. The tree scores above 0.85 and the majority baseline scores about a third, because the three species are equally common. plot_tree draws the questions: the top box asks about a petal measurement, the same kind of question as your hand rule. The classes are coded 0, 1 and 2 for setosa, versicolor and virginica. The codes name categories, so species 2 is not twice species 1. Accuracy is correct predictions divided by all predictions.

Investigate.

1. What does the top box of the drawn tree ask?
2. Why does the baseline score near one third?
3. Why is species code 2 not twice species code 1?

Modify. Change max_depth to 1 and run again.

Block B. Back to the hand rule

Score your session 4 rule on all 150 flowers. Petal length is column 2 of the data. The rule says setosa below 2.5, versicolor below 4.8 and virginica otherwise.

Predict. Predict whether your hand rule will score above or below the tree. Write your guess on paper first.

Your guess, written before you run it

```
lengths = iris.data[:, 2]
guess = (lengths >= 2.5).astype(int) + (lengths >= 4.8).astype(int)
hand_acc = (guess == iris.target).mean()
print(round(hand_acc, 3))
```

Run it, then compare. The hand rule scores above 0.9 on all 150 flowers. Each comparison gives True or False for every flower and astype(int) turns them into 1 and 0. Adding the two gives 0 for setosa, 1 for versicolor and 2 for virginica. Your rule and the tree use the same idea. The tree chose its question and cut-off by itself, and you chose yours by looking at eight flowers.

Investigate.

1. Why does the hand rule use only petal length?

2. Whose score is fairer: the hand rule on all 150 flowers, or the tree on the 38 held-out flowers of a 112 and 38 split?

Modify. Change 4.8 to 5.0 in the rule and score it again.

Make

Convert both accuracies to percentages. Name the target and one input. Explain why species code 2 is not twice species code 1.

Save your work as ML16. Keep the worked example and put your own attempt below it. Write two sentences: what you changed and how you checked it.

Error card

You see	It means	Fix
ValueError: could not convert string to float	A category name was passed where numbers are needed.	Use iris.target for the labels and iris.data for the inputs.
NameError: name 'iris' is not defined	The cell that loads iris has not run.	Run the cells in order from the top.

Exit ticket

A tree learns its questions from examples and is compared against a baseline and against a rule I wrote.

AI use follows AI_USE_POLICY.pdf. In weeks 1 to 6 no AI-generated solutions are used for exercises, labs or build tasks. Practise for 20 to 30 minutes between classes by repeating the Make task with one value changed. If you are stuck for 30 minutes, save the error and bring it to class. A file saved on your laptop has not been shared until you use the route in START_HERE.pdf.